1 Logarithmic functions and indices

What students need to learn	Notes
A The functions a^x and $\log_b x$ (where b is a natural number greater than one)	A knowledge of the shape of the graphs of a^x and $\log_b x$ is expected, but not a formal expression for the gradient.
B Use and properties of indices and logarithms, including change of base	To include: $\log_a xy = \log_a x + \log_a y$, $\log_a \frac{x}{y} = \log_a x - \log_a y$, $\log_a x^k = k \log_a x$, $\log_a a = 1$ $\log_a 1 = 0$ The solution of equations of the form $a^x = b$. Students may use the change of base formulae: $\log_a x = \frac{\log_b x}{\log_b a}$ $\log_a b = \frac{1}{\log_b a}$
C Simple manipulation of surds	Students should understand what surds represent and their use for exact answers. Manipulation will be very simple. For example: $5\sqrt{3} + 2\sqrt{3} = 7\sqrt{3}$ $\sqrt{48} = 4\sqrt{3}$
D Rationalising the denominator	$10 \times \frac{1}{\sqrt{5}} = 2\sqrt{5}$ or $\frac{1}{2-\sqrt{3}}$